

Section A (Attempt any two questions. Each question carries 6 marks)

Q1. Find Laplace transform of $e^{-4t} \int_0^t t \sin^2 t dt$

Q2. Using Laplace transform evaluate $\int_0^\infty \left(\frac{e^{-t} - e^{-2t}}{t} \right) dt$.

Q3. Find inverse Laplace transform of $\frac{1}{(s-2)(s+2)^2}$

Q4. Find the Fourier integral representation of the function $f(x) = \begin{cases} e^{ax} & x \leq 0 \\ e^{-ax} & x \geq 0 \end{cases}$ for $a > 0$

Hence show that $\int_0^\infty \frac{\cos \lambda x}{\lambda^2 + a^2} d\lambda = \frac{\pi}{2a} e^{-ax}$, $x \geq 0$.

Section B (Attempt any one question. Each question carries 8 marks)

Q1. Express the function $f(x) = \begin{cases} 1 & \text{when } |x| \leq 1 \\ 0 & \text{when } |x| > 1 \end{cases}$ as Fourier integral. Hence, evaluate $\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$.

Q2. Use Laplace transform to solve the differential equation $y'' - 3y' + 2y = 4$, $y(0) = 2$, $y'(0) = 3$.

Section A (Attempt any two questions. Each question carries 6 marks)

Q1. Show that Z transform of $\frac{1}{n!} = e^{\frac{1}{z}}$ hence evaluate Z transform of $\frac{1}{(n+1)!}$ and $\frac{1}{(n+2)!}$.

Q2. Find Z transform of (i) $n \sin n\theta$ (ii) $a^n n^2$

Q3. Use iteration method to approximate the real root of $\cos x = 3x - 1$, correct to 3 decimal places if the root $\in (0,1)$.

Q4. Use Newton Raphson method to approximate the real root of $x^5 - 5x^2 + 3 = 0$, correct to 4 decimal places.

Section B (Attempt any one question. Each question carries 8 marks)

Q1. Use Z transform to solve $y_{n+2} + 4y_{n+1} + 3y_n = 3^n$, given $y_0 = 0, y_1 = 1$.

Q2. Use Bisection method to find a real root of the equation $x^3 - 4x - 9 = 0$ correct to 3 decimal places if the root $\in (2,3)$.